Hints: Inequalities – 1. The Cauchy–Bunyakovsky inequality as a technique

Homework

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Split $f = f_e + f_o$ into its even and odd parts. In $L^2[-1,1]$ they are orthogonal, so $\int f^2 = \int f_e^2 + \int f_o^2$, while $\int f = \int f_e$ and $\int xf = \int xf_o$.

Hint 2. The functions $e_0 = \frac{1}{\sqrt{2}}$ and $e_1 = \sqrt{\frac{3}{2}}\,x$ have norm 1. By Cauchy–Bunyakovsky (Bessel), $\int f_e^2 \geqslant \langle f_e, e_0\rangle^2 = \frac{1}{2}\big(\int f\big)^2$ and $\int f_o^2 \geqslant \langle f_o, e_1\rangle^2 = \frac{3}{2}\big(\int xf\big)^2$. Add.

Hint 3. Equality holds iff f_e is proportional to e_0 and f_o to e_1, i.e. exactly for $f(x) = a + bx$.

Problem 2

Hint 1. Put $g(x) = \int_0^1 f(x,y)\,dy$, $h(y) = \int_0^1 f(x,y)\,dx$ and $I = \int_0^1\int_0^1 f$. The inequality reads $\int h^2 + \int g^2 \leqslant I^2 + \iint f^2$. Look for a square whose integral is the difference.

Hint 2. Expand $\iint \big(f(x,y) - g(x) - h(y) + I\big)^2\,dx\,dy \geqslant 0$. Every cross term reduces to $\int g^2$, $\int h^2$ or I^2 (e.g. $\iint fg = \int g^2$, $\iint gh = I^2$), and the total is exactly $\iint f^2 - \int g^2 - \int h^2 + I^2$.

Problem 3

Hint 1. First show that $\mathrm{Var}(f) = \min_c \int_0^1 (f-c)^2$: expand $\int (f - \mu(f) + \mu(f) - c)^2$; the cross term vanishes.

Hint 2. Hence $\mathrm{Var}(fg) \leqslant \int_0^1 (fg - \mu(f)\mu(g))^2$. Split $fg - \mu(f)\mu(g) = (f-\mu(f))g + (g-\mu(g))\mu(f)$.

Hint 3. Use $(u+v)^2 \leqslant 2u^2 + 2v^2$ pointwise, then $|g| \leqslant M(g)$ and $|\mu(f)| \leqslant M(f)$.

Problem 4

Hint 1. Split $\int_0^1 f = \int_0^{1/2} f + \int_{1/2}^1 f$ and integrate each part by parts using $f(\frac{1}{2}) = 0$, with x on the left half and $x-1$ on the right half.

Hint 2. You get $\int_0^{1/2} f = -\int_0^{1/2} xf'$ and $\int_{1/2}^1 f = -\int_{1/2}^1 (x-1)f'$, i.e. $\int_0^1 f = -\int_0^1 k(x)f'(x)\,dx$ with $k(x) = x$ on $[0, \frac{1}{2}]$ and $x-1$ on $(\frac{1}{2}, 1]$.

Hint 3. Apply Cauchy–Bunyakovsky: $\big(\int_0^1 f\big)^2 \leqslant \int_0^1 k^2 \cdot \int_0^1 (f')^2$ and $\int_0^1 k^2 = 2\int_0^{1/2} x^2\,dx = \frac{1}{12}$.

Problem 5

Hint 1. Integrate $\int_0^1 xf$ by parts with $f(1) = 0$ so that xf' appears.

Hint 2. $\int_0^1 xf\,dx = \big[\frac{x^2}{2}f\big]_0^1 - \frac{1}{2}\int_0^1 x^2 f'\,dx$, so $2\int_0^1 xf = -\int_0^1 x \cdot xf'(x)\,dx$.

Hint 3. Cauchy–Bunyakovsky with $\int_0^1 x^2 = \frac{1}{3}$: $4\big(\int xf\big)^2 \leqslant \frac{1}{3}\int (xf')^2$. Compare with Problem 4: there $\int f = -\int kf'$ with the kernel k = signed distance to the zero $\frac{1}{2}$ and $\int k^2 = \frac{1}{12}$; here the zero is at 1, the derivative carries the weight x, and the constant $12 = 4 \cdot 3$ comes from the factor 2 of the integration by parts and $\int x^2 = \frac{1}{3}$.

Problem 6

(a)

Hint 1. Square the right-hand side and keep, for each pair $j<i$, the cross term $2x_ix_j/\sqrt{ij}$; use $x_j \geqslant x_i$ for $j \leqslant i$ to replace it by something with x_i^2.

Hint 2. You get $\big(\sum \frac{x_i}{\sqrt{i}}\big)^2 \geqslant \sum_i x_i^2\Big(\frac{2}{\sqrt{i}}\sum_{j\leqslant i}\frac{1}{\sqrt{j}} - \frac{1}{i}\Big)$, and $\sum_{j\leqslant i}\frac{1}{\sqrt{j}} \geqslant \frac{i}{\sqrt{i}} = \sqrt{i}$, so each bracket is at least $2-\frac{1}{i} \geqslant 1$.

(b)

Hint 1. Apply (a) to the tail sequence $x_m, x_{m+1}, \ldots$: $\big(\sum_{i\geqslant m} x_i^2\big)^{1/2} \leqslant \sum_{i\geqslant m}\frac{x_i}{\sqrt{i-m+1}}$.

Hint 2. Swap the sums: the left side is at most $\sum_i x_i \sum_{m=1}^{i}\frac{1}{\sqrt{m(i+1-m)}}$. The inner sum is bounded: by symmetry it is at most $2\sqrt{\frac{2}{i+1}}\sum_{m\leqslant(i+1)/2}\frac{1}{\sqrt{m}} \leqslant 4$ (it is a Riemann sum for $\int_0^1 \frac{dt}{\sqrt{t(1-t)}} = \pi$). So $C=4$ works.

Problem 7

Hint 1. Write the product as a double integral: $\int e^f \int e^{-f} = \int_0^1\int_0^1 e^{f(x)-f(y)}\,dx\,dy$, and symmetrise in $x \leftrightarrow y$.

Hint 2. This equals $\iint \cosh\big(f(x)-f(y)\big)$, and $\cosh t \geqslant 1+\frac{t^2}{2}$. Finally $\frac{1}{2}\iint (f(x)-f(y))^2 = \int f^2 - \big(\int f\big)^2$.

Problem 8

Hint 1. The point is that $\int_0^T \ln f(x)\,dx = \int_0^T \ln f(x+\alpha)\,dx$ by periodicity.

Hint 2. Use $t \geqslant 1+\ln t$ with $t = \frac{f(x)}{f(x+\alpha)}$ and integrate over a period: $\int_0^T \frac{f(x)}{f(x+\alpha)} \geqslant T + \int_0^T \big(\ln f(x) - \ln f(x+\alpha)\big) = T$.

Problem 9

Hint 1. For $v_1,\ldots,v_m$ and real c_i, $\sum_{i,j} c_ic_j\langle v_i, v_j\rangle = \big\|\sum c_iv_i\big\|^2 \geqslant 0$: the Gram matrix G is positive semidefinite, so $\det G \geqslant 0$.

Hint 2. For $m=2$, $\det G = \|v_1\|^2\|v_2\|^2 - \langle v_1, v_2\rangle^2 \geqslant 0$ is the Cauchy–Bunyakovsky inequality, with equality iff v_1, v_2 are dependent.

Hint 3. Expressing the v_i in an orthonormal basis of their span, $G = V^TV$ and $\det G = (\det V)^2$ when the dimension is m: $\det G$ is the squared m-dimensional volume of the parallelepiped spanned by the v_i (zero iff they are dependent).

Problem 10

Note. As printed (with products) the inequality is false: for $f \equiv c \in (0,1)$ it becomes $c^2(c^2+1) < c^3$. The original SEEMOUS 2018 problem has quotients: $\dfrac{2\int_0^1 xf^2}{\int_0^1(f^2+1)} < \dfrac{\int_0^1 f^2}{\int_0^1 f}$, i.e. $2\int xf^2 \cdot \int f < \int f^2 \cdot \int(f^2+1)$. The hints follow this version.

Hint 1. Compare the factors separately: which of $\int xf^2$ and $\int f^2$ is larger, and which of $2\int f$ and $\int(1+f^2)$?

Hint 2. $\int_0^1 xf^2 \leqslant \int_0^1 f^2$ because $x \leqslant 1$, and $1+f^2-2f = (1-f)^2 > 0$ everywhere, so $\int(1+f^2) > 2\int f$ (a positive Riemann integrable function has positive integral). Multiply, using $\int f^2 > 0$ and $\int f > 0$.